Akib Javed, Ph.D.

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Education

Academic Degrees

2017 – 2024 **Ph.D.** in **Cartography and Geographical Information Systems** from the **State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS)**, Wuhan University, China.

2014 – 2015 **M.S.** in **Geography and Environmental Studies** from University of Rajshahi.

2010 – 2013 **B.Sc.** in **Geography and Environmental Studies** from University of Rajshahi.

Scholarship

2017 **CSC Scholarship**: Awarded by Chinese Scholarship Council for doctoral studies

Employment History

February 2025 – Present **GIS Analyst.** Georeferenced and digitized old cadastral maps for a national land zoning project, improving land management efficiency for government agencies and reducing manual processing time by 10% within my teams.

June 2025 – August 2025 **GIS Supervisor (Part-time).** Led a GIS team of 3 analysts, improving project delivery speed by 20% and ensuring 95% accuracy in all spatial data outputs.

June 2019 – June 2019 **Internship.** Participated in week long internship program and familiarised with latest geospatial tools and softwares at Hi-TARGET SURVEYING INSTRUMENT Co. Ltd, Guangzhou, China from 2019-06-24 to 2019-06-28..

July 2017 – July 2017 **Visiting Lecturer.** Taught GIS and remote sensing courses to 100+ undergraduate students, improving average student geospatial knowledge enthusiast by 5% through hands-on session within a week.

April 2017 – August 2017 **Research Assistant.** Analyzed land use patterns for the Adivasi People project, generating key insights that informed community development strategies for 3,000+ beneficiaries.

July 2016 – March 2017 **Research Assistant (Part-time).** Supported PhD research by analyzing primary data, accelerating data processing time by 35% and contributing to 5 peer-reviewed publications on environmental science over three years.

March 2016 – June 2016 **Research Assistant (Part-time).** Assisted Prof. Dr. Raquib Ahmed (Rajshahi University) in environmental change research, utilizing geospatial analysis to increase data accuracy by 10%.

January 2011 – June 2015 **Journalist (Part-time).** Published 30+ news articles as a university correspondent, while working at 'Daily Janata'.

Research

Academic Service

Journal Peer Review — Remote Sensing of Environment, Advances in Space Research, Journal of Selected Topics in Applied Earth Observations and Remote Sensing, City and Environment Interactions

Conference Review — ISPRS Geospatial Week 2025, Dubai

Research Co-supervision

2025 — Spatio-temporal Mapping of Forest Areas of Bangladesh: A remote sensing base study using Google Earth Engine (GEE) platform.

2024 — Spatiotemporal Analysis of Droughts in Northern Bangladesh using MODIS Time Series (2000-2023).

Spatiotemporal Mapping of Urban Air Pollutants: A Comparative Study of Bangladesh.

Research Experience

M.Sc. project — Analyzed land use changes in Teknaf municipality and it's surroundings, show change detection over five years, identifying critical deforestation trends and propose policy for 50,000+ residents.

B.Sc. project — Investigated urban transformation in Rajshahi City, revealing key patterns of urban sprawl driven by road networks and informing sustainable development strategies for 500,000+ residents.

2017 — Conducted intensive land study for the Adivasi People project, collecting data from 300+ households and informing community development strategies for 5,000+ beneficiaries.

2013 — Authored a detailed field reports from a socioeconomic survey on Ashariadaha Island, providing critical data that informed local lifestyle and economic activities for 3,000+ residents.

2012 — Generated a comprehensive field reports from a socioeconomic survey on Dubla Char Island, delivering key insights of local fisherman.

2011 — Authored a field reports on industrial regions of Kushtia, providing critical data that informed major economic activities for four factories.

Research Publications

Journal Articles

1. Muhammad Nasar Ahmad, Xiaohui Yuan, Lichuan Gu, Shuai Yang, Rana Waqar Aslam, **Akib Javed**, and George D. Bathrellos. "Advancements in deep learning methods for precise weed and rice classification from UAV imagery: a comprehensive review". In: *Discover Agriculture* 4.1 (Jan. 2026), p. 8. ISSN: 2731-9598. DOI: 10.1007/s44279-025-00469-0.

2. Muhammad Nasar Ahmad, Fahad Almutlaq, Md. Enamul Huq, Fakhrul Islam, **Akib Javed**, Hariklia D. Skilodimou, and George D. Bathrellos. "Enhanced urban impervious surface land use mapping using a novel multi-sensor feature fusion method and remote sensing data". In: *Environmental Earth Sciences* 84.9 (Apr. 2025), p. 228. ISSN: 1866-6299. DOI: 10.1007/s12665-025-12217-0.

3. Muhammad Nasar Ahmad, Hariklia D. Skilodimou, Fakhrul Islam, **Akib Javed**, and George D. Bathrellos. "Drainage Network Generation for Urban Pluvial Flooding (UPF) Using Generative Adversarial Networks (GANs) and GIS Data". In: *Sustainability* 10 (May 2025). ISSN: 2071-1050. DOI: 10.3390/su17104380.

4. Iffat Ara, Rubaiatul Islam Zerin, Laila Anjum Eva, Md. Kamrul Hossain, Mahmuda Hossain Mou, **Akib Javed**, Shahin Ali, Md. Anisul Kabir, and Md Masud Parves Rana. "Comparative Hydrogeochemical

Characterization and Environmental Controls on Groundwater Quality, Irrigation Suitability, and Health Risk across Northern and Coastal Hydrological Settings of Bangladesh". In: *Cleaner Water* (Dec. 2025), p. 100194. ISSN: 2950-2632. DOI: https://doi.org/10.1016/j.clwat.2025.100194.

5 Rubaiatul Islam Zerin, Iffat Ara, Shahin Ali, Md. Kamrul Hossain, **Akib Javed**, Shamsun Nahar Ratna, and Md. Anisul Kabir. "Geostatistical modeling of soil physicochemical properties and environmental drivers in agro-industrial landuse systems in Bangladesh". In: *Geomatica* 77.2 (Sept. 2025), p. 100075. ISSN: 1195-1036. DOI: https://doi.org/10.1016/j.geomat.2025.100075.

6 M. N. Ahmad, Zhenfeng Shao, **Akib Javed**, Israr Ahmad, Fakhrul Islam, Hariklia D. Skilodimou, and George D. Bathrellos. "Optical-SAR Data Fusion Based on Simple Layer Stacking and the XGBoost Algorithm to Extract Urban Impervious Surfaces in Global Alpha Cities". In: *Remote Sensing* 16.5 (2024). DOI: 10.3390/rs16050873.

7 Muhammad Nasar Ahmad, Zhenfeng Shao, Xiongwu Xiao, Peng Fu, **Akib Javed**, and Iffat Ara. "A novel ensemble learning approach to extract urban impervious surface based on machine learning algorithms using SAR and optical data". In: *International Journal of Applied Earth Observation and Geoinformation* 132 (2024), p. 104013. ISSN: 1569-8432. DOI: https://doi.org/10.1016/j.jag.2024.104013.

8 Md. Enamul Huq, Xijun Wu, **Akib Javed**, Ying Dong, Bingbing Li, David Lopez-Carr, Jiang Wu, Jing Liu, Yaning Zhang, Fanping Zhang, Muhammad Riaz, Md. Nazirul Islam Sarker, Walid Soufan, Khalid F. Almutairi, and Aqil Tariq. "Changes monitoring in Hongjiannao Lake from 1987 to 2023 using Google Earth Engine and analysis of climatic and anthropogenic forces". In: *Physics and Chemistry of the Earth, Parts A/B/C* 136 (2024), p. 103756. DOI: https://doi.org/10.1016/j.pce.2024.103756.

9 **Akib Javed**, Zhenfeng Shao, Iffat Ara, M. N. Ahmad, Enamul Huq, Nayyer Saleem, and Fazlul Karim. "Development of Soil-Suppressed Impervious Surface Area Index for Automatic Urban Mapping". In: *Photogrammetric Engineering & Remote Sensing* 90.1 (2024), pp. 33–43. ISSN: 0099-1112. DOI: 10.14358/PERS.23-00043R2.

10 Zhenfeng Shao, M. N. Ahmad, and **Akib Javed**. "Comparison of Random Forest and XGBoost Classifiers Using Integrated Optical and SAR Features for Mapping Urban Impervious Surface". In: *Remote Sensing* 16.4 (2024), p. 665. ISSN: 2072-4292. URL: https://www.mdpi.com/2072-4292/16/4/665.

11 M. N. Ahmad, Z. Shao, and **Akib Javed**. "Modelling land use/land cover (LULC) change dynamics, future prospects, and its environmental impacts based on geospatial data models and remote sensing data". In: *Environmental Science and Pollution Research* 30.12 (2023), pp. 32985–33001. ISSN: 1614-7499 (Electronic) 0944-1344 (Linking). DOI: 10.1007/s11356-022-24442-2.

12 M. N. Ahmad, Zhenfeng Shao, and **Akib Javed**. "Mapping impervious surface area increase and urban pluvial flooding using Sentinel Application Platform (SNAP) and remote sensing data". In: *Environmental Science and Pollution Research* 30.60 (2023), pp. 125741–125758. ISSN: 1614-7499. DOI: 10.1007/s11356-023-30990-y.

13 M. N. Ahmad, Zhenfeng Shao, **Akib Javed**, Fakhrul Islam, Hafiz Haroon Ahmad, and Rana Waqar Aslam. "The Cellular Automata Approach in Dynamic Modelling of Land Use Change Detection and Future Simulations Based on Remote Sensing Data in Lahore Pakistan". In: *Photogrammetric Engineering & Remote Sensing* 89.1 (2023), pp. 47–55. ISSN: 0099-1112. DOI: 10.14358/pers.22-00102r2.

14 M. Y. Ali, M. Shahrier, A. A. Kafy, I. Ara, **Akib Javed**, M. A. Fattah, Z. A. Rahaman, and K. Tripura. "Environmental impact assessment of tobacco farming in northern Bangladesh". In: *Heliyon* 9.3 (2023), e14505. ISSN: 2405-8440 (Print) 2405-8440 (Electronic) 2405-8440 (Linking). DOI: 10.1016/j.heliyon.2023.e14505.

15 **Akib Javed**, Z. F. Shao, B. Bai, Z. Y. Yu, J. B. Wang, I. Ara, M. E. Huq, M. Y. Ali, N. Saleem, M. N. Ahmad, N. S. Sumari, and Mardia. "Development of normalized soil area index for urban studies using remote sensing data". In: *Geofizika* 40.1 (2023), pp. 29–49. ISSN: 0352-3659. DOI: 10.15233/gfz.2023.40.2.

16 Zhenfeng Shao, M. N. Ahmad, **Akib Javed**, Fakhrul Islam, Zahid Jahangir, and Israr Ahmad. "Expansion of Urban Impervious Surfaces in Lahore (1993-2022) Based on GEE and Remote Sensing Data". In:

Photogrammetric Engineering and Remote Sensing 89.8 (2023), pp. 479–486. ISSN: 0099-1112. DOI: 10.14358/pers.23-00001r2.

17. M. N. Ahmad, Z. F. Shao, Andaleeb Yaseen, Muhammad Nabeel Khalid, and **Akib Javed**. "The Simulation and Prediction of Land Surface Temperature Based on SCP and CA-ANN Models Using Remote Sensing Data: A Case Study of Lahore". In: *Photogrammetric Engineering and Remote Sensing* 88.12 (2022), pp. 783–790. ISSN: 0099-1112. DOI: 10.14358/pers.22-00071r2.

18. Md Yeamin Ali, Abdulla-Al Kafy, Zullyadini A Rahaman, Md Fakrul Islam, Md Redwanur Rahman, Iffat Ara, Mst Rupali Akhtar, and **Akib Javed**. "Comparative occupational health risk between tobacco and paddy farming people in Bangladesh". In: *SSM-Mental Health* 2 (2022), p. 6.

19. **Akib Javed**, Q. M. Cheng, H. Peng, O. Altan, Y. Li, I. Ara, E. Huq, Y. Ali, and N. Saleem. "Review of Spectral Indices for Urban Remote Sensing". In: *Photogrammetric Engineering and Remote Sensing* 87.7 (2021), pp. 513–524. ISSN: 0099-1112. DOI: 10.14358/Pers.87.7.513.

20. M. E. Huq, Q. M. Cheng, O. Altan, A. Z. M. Shoeb, M. A. Hossain, M. N. I. Sarker, N. Saleem, **Akib Javed**, X. Long, A. A. Al Dughairi, M. M. P. Rana, A. Al Mamun, and M. M. Rahman. "Assessing vulnerability for inhabitants of Dhaka City considering flood-hazard exposure". In: *Geofizika* 37.2 (2020), pp. 97–130. DOI: 10.15233/gfz.2020.37.5. WOS: 000608071700001. (Visited on 10/07/2025).

21. Nayyer Saleem, Md. Enamul Huq, Nana Yaw Danquah Twumasi, **Akib Javed**, and Asif Sajjad. "Parameters Derived from and/or Used with Digital Elevation Models (DEMs) for Landslide Susceptibility Mapping and Landslide Risk Assessment: A Review". In: *ISPRS International Journal of Geo-Information* 8.12 (2019), pp. 545–572.

Books and Chapters

1. Md Enamul Huq, David Lopez-Carr, Aqil Tariq, Xijun Wu, Jiang Wu, Ying Dong, Zawawi Daud, **Akib Javed**, Md Nazirul Islam Sarker, KM Mehedi Adnan, et al. "Advancements in remote sensing and GIS for water resource management and SDG monitoring". In: *Utilizing Earth Observation Data in Reaching Sustainable Development Goals*. Elsevier, 2026, pp. 189–218. DOI: https://doi.org/10.1016/B978-0-443-30204-6.00007-3.

2. Md Enamul Huq, Zhenfeng Shao, Ahmed Abdullah Al Dughairi, Md Nazirul Islam Sarker, Cai Bowen, Abdullah Al Mamun, Nayyer Saleem, **Akib Javed**, and Md Mahabubur Rahman. "Measuring Vulnerability to Flash Flood of Urban Dwellers". In: *Wadi Flash Floods*. Springer, Singapore, 2022, p. 317.

3. Md. Enamul Huq, A. Z. M. Shoeb, Mallik Akram Hossain, Shah Fahad, M. M. Kamruzzaman, **Akib Javed**, Nayyer Saleem, K. M. Mehedi Adnan, Swati Anindita Sarker, and Md Yeamin Ali. "Measuring Vulnerability to Environmental Hazards: Qualitative to Quantitative". In: *Environment, Climate, Plant and Vegetation Growth*. Springer, 2020, pp. 421–452.

Conference Proceedings

1. Most. Sajeda Akter, Iffat Ara, and **Akib Javed**. "A Comprehensive Evaluation of Environmental Literacy: A Case Study in Kushtia Municipality, Bangladesh". In: *ABSTRACTS BOOK OF FIRST INTERNATIONAL CONFERENCE ON BANGLADESH STUDIES (ICBS-I) 2025*. Ed. by Osman AKGÜL, Hafijur Rahman, and ASM Mahmudul Hasan. Presented via online. Istanbul Ticaret University, Türkiye: Istanbul Ticaret University, Türkiye, Apr. 2025, p. 24.

2. Most. Sajeda Akter, Iffat Ara, and **Akib Javed**. "A Comprehensive Evaluation of Environmental Literacy: A Case Study in Kushtia Municipality, Bangladesh". In: *Conference Proceedings on 1st International Conference on Bangladesh Studies (ICBS-I), 2025*. Hybrid conference held on April 19-20, 2025. Istanbul Ticaret University, Türkiye, Apr. 2025. URL: https://cpsrinfo.org/.

3. Shahin Ali, **Akib Javed**, and Iffat Ara. "Agricultural Drought in Northern Bangladesh: A Spatiotemporal Analysis Using MODIS Time Series From 2001 To 2023". In: *ABSTRACTS BOOK OF FIRST INTERNATIONAL CONFERENCE ON BANGLADESH STUDIES (ICBS-I) 2025*. Ed. by Osman AKGÜL, Hafijur Rahman, and ASM Mahmudul Hasan. Presented via online. Istanbul Ticaret University, Türkiye: Istanbul Ticaret University, Türkiye, Apr. 2025, p. 46.

4. Laila Anjum Eva, Iffat Ara, Md. Shohel Khan Khan, Md. Aminur Rahman, and **Akib Javed**. "From Resource to Risk: Evaluating the Rainwater Quality and Associated Health Issues in the Southern Coastal Bangladesh". In: *Conference Proceedings on 10th International Conference on Water and Flood Management - ICWFM 2025*. Vol. 10. Institute of Water and Flood Management, BUET, Feb. 2025, pp. 277–284.

5. Faria Islam, Iffat Ara, Md. Shohel Khan, and **Akib Javed**. "Assessment of Particulate Matter Exposure and Associated Human Health Risk: Comparison among Non-automated, Automated and Semi-automated Rice Processing Industry Workers in Bangladesh". In: *Proceedings of the 1st International Conference on Geosciences for Achieving Sustainable Development Goals (ICGSDG-2025)*. Faculty of Geosciences, University of Rajshahi, Bangladesh, Nov. 2025, p. 162.

6. Humaira Rashid, **Akib Javed**, and Iffat Ara. "A Remote Sensing Assessment of Air Pollution in Five Bangladeshi Industrial Cities". In: *INTERNATIONAL GLOBAL SUSTAINABILITY AND DEVELOPMENT CONGRESS Full Text Book*. Ed. by Oktay Özkan and Şükrü Taner Azgin. New York, NY: Liberty Publishing House, Nov. 2025, pp. 873–897. ISBN: 979-8-89695-250-3. URL: iksadkongre.com/_files/ugd/614b1f_c112d6b2a09442749bd2d5579d9cdb53.pdf.

7. Humaira Rashid, **Akib Javed**, and Iffat Ara. "A Remote Sensing Assessment of Air Pollution in Five Bangladeshi Industrial Cities". In: *Global Sustainability and Development Congress Abstract Book*. Ed. by Oktay Özkan and Şükrü Taner Azgın. Kayseri, Türkiye: Liberty Academic Publishers, Oct. 2025, p. 245. ISBN: 979-8-89695-212-1.

8. Rafia Razzak Ria, Iffat Ara, and **Akib Javed**. "An Analysis of Governance Framework Effectiveness and Internal Waste Management Practices in The Hospitals of Rangpur Sadar Upazila, Bangladesh". In: *ABSTRACTS BOOK OF FIRST INTERNATIONAL CONFERENCE ON BANGLADESH STUDIES (ICBS-I) 2025*. Ed. by Osman AKGÜL, Hafijur Rahman, and ASM Mahmudul Hasan. Presented via online. Istanbul Ticaret University, Türkiye: Istanbul Ticaret University, Türkiye, Apr. 2025, p. 77.

9. Rafia Razzak Ria, Iffat Ara, and **Akib Javed**. "An Analysis of Governance Framework Effectiveness and Internal Waste Management Practices in the Hospitals of Rangpur Sadar Upazila, Bangladesh". In: *Conference Proceedings on 1st International Conference on Bangladesh Studies (ICBS-I), 2025*. Hybrid conference held on April 19-20, 2025. Istanbul Ticaret University, Türkiye, Apr. 2025. URL: https://cpsrinfo.org/.

10. Rowshonul Islam Zerin, Iffat Ara, Laika Anjum, Mahmudul Hossain Moti, Md. Kamrul Hossain, and **Akib Javed**. "Index-based Assessment of Groundwater Quality and Suitability for Irrigation Across Major Agricultural Regions of Bangladesh". In: *Proceedings of the 1st International Conference on Geosciences for Achieving Sustainable Development Goals (ICGSDG-2025)*. Faculty of Geosciences, University of Rajshahi, Bangladesh, Nov. 2025, p. 109.

11. ME Huq, Shao Zhenfeng, Cai Bowen, Nayyer Saleem, Ahmed Abdullah Aldughairi, Danquan Twumasi Nana Yaw, **Akib Javed**, and Md. Mahabubur Rahman. "Measuring Vulnerability to Flash Flood for Urban Inhabitants: A Case Study of Dhaka Megacity, Bangladesh". In: *The Fifth International Symposium on Flash Floods in Wadi Systems (ISFF2020)*. 2020, p. 87.

12. Md Enamul Huq, AZM Shoeb, **Akib Javed**, Zhenfeng Shao, Mallik Akram Hossain, and Most Sinthia Sarven. "Measuring vulnerability for city dwellers exposed to flood hazard: A case study of Dhaka City, Bangladesh". In: *Urban Intelligence and Applications: Proceedings of ICUIA 2019*. Springer International Publishing, 2019, pp. 207–215. ISBN: 3030450988.

13. M. Yeamin Ali, Mehadi Hassan, M. Atiqur Rahmana, Abdulla-AI Kafy, Iffat Ara, **Akib Javed**, and M. Redwanur Rahman. "Life cycle energy and cost analysis of small scale biogas plant and solar PV system in

rural areas of Bangladesh". In: *2nd International Conference on Energy and Power, ICEP2018, 13–15 December 2018, Sydney, Australia*. Vol. 160. Energy Procedia, ELSEVIER, 2018, pp. 277–284.

14 **Akib Javed** and Shao Zhenfeng. "Smart Application using GNSS, GIS and Remote Sensing: Flood Prediction and Monitoring". In: *First International Workshop on Artificial Intelligence and Big Data Analysis 2017*. 2017, p. 67.

15 **Akib Javed** and Raquib Ahmed. "Philosophy of Geography Education: Some Perspectives and Progress in Bangladesh Context". In: *International Conference on Geography Education 2015*. 2015, p. 33.

Posters

1 **Akib Javed** and M. N. Ahmad. "Chinese Desert Reclamation: Turning Barren Land into Green Oases". In: (Mar. 2023). DOI: 10.13140/RG.2.2.30223.46240.

2 **Akib Javed**, Zhenfeng Shao, Iffat Ara, M. N. Ahmad, Enamul Huq, Nayyer Saleem, and Fazlul Karim. "Soil Suppressed Generalized Impervious Surface Area Index (GISAI) for Automatic Megacity Mapping". In: (Mar. 2023). DOI: 10.13140/RG.2.2.35676.05761.

Seminars, Conferences, Workshops, Trainings

Seminars

2023-10-12 **Mobile Mapping Systems: Demonstrated Technology that Keeps Evolving** by Prof. Ayman F. Habib, at LIESMARS, Wuhan University.

2023-10-03 **Unleashing the Power of Words: Writing Practices and Processes** by Stephen C. McClure, at LIESMARS, Wuhan University.

2023-06-20 **Smart City** by Prof. Dr. Zhenfeng Shao, at LIESMARS, Wuhan University.

2023-05-30 **Navigating the Academic Landscape: Insights and Strategies for Successful Research and Career Development** by Prof. Naser El-Sheimy, at LIESMARS, Wuhan University.

2023-05-21 **Forest Aboveground Biomass Estimation with Airborne LiDAR data** by Prof. Dengsheng Lu, at LIESMARS, Wuhan University.

2023-05-06 **LEO Navigation Augmentation System from Luojia-1A/2A Satellite - Design, Progress and Application** by Prof. Lei Wang, at LIESMARS, Wuhan University.

Conferences

17-19 November 2024 Participated in the **2024 International Symposium on Spatio-Temporal Intelligence and 2024 International Conference on Geospatial Information Science: Education, Innovation and Applications** organized by LIESMARS, Wuhan University.

15-16 October 2023 Participated in the **International Conference on Geospatial Information Science: Education, Innovation and Applications** organized by LIESMARS, Wuhan University.

16-18 October 2018 Participated in the **International Conference on Advance Remote Sensing 2018** organized by LIESMARS, Wuhan University.

20 December 2017 Participated in the **International Symposium on key technologies and applications of Video Surveillance and Intelligent Analysis in Hengqin Island** organized by Zhuhai City Development Authority, Guangzhou, Guangdong, China.

11-13 September 2015 Participated in the **International Conference on Geography Education 2015** organized by the Dept. of Geography and Environmental Studies, University of Rajshahi, Bangladesh and Institute of Geography, University of Freiburg, Germany at the University of Rajshahi, Bangladesh.

Seminars, Conferences, Workshops, Trainings (continued)

07-08 March 2014 — Participated in the **International Conference on Environment and Development** organized by the Department of Geography and Environmental Studies, University of Rajshahi, supported by Higher Education Quality Enhancement Project (HEQEP).

Workshop

2019 — Ten days long Workshop title **Zhuhai One Hyperspectral Remote Sensing Training** organized by Zhuhai ORBITA Aerospace Science & technology Co., Ltd. at Obit Science and Technology Park, No. 1 Baisha Road, Tangjia East Bank, Zhuhai City, China.

2017 — Three days long workshop title **The 1st LIESMARS International Graduate Workshop on GeoInformatics** organized by LIESMARS, Wuhan University, Wuhan City, China (November 25-27, 2017).

2014 — Participated to a Workshop on **Teaching Method and Curriculum Development** at the Department of Geography and Environmental Studies, University of Rajshahi, Bangladesh: Organized by University of Education Freiburg, Germany and University of Rajshahi, Bangladesh (November 10-12, 2014).

2013 — Participated on Teacher Training Workshop on **Innovative Teaching Methodology, Learning Strategies and Didactics** at the University of Rajshahi, Bangladesh. Organized by the University of Education Freiburg, Germany and the University of Rajshahi, Bangladesh (18-20 February 2013).

Training and Summer School

2024 — Completed ten-day long training Program titled **Application of Mathematical Modelling & GIS Technology for Feasibility Study of Water Development Projects** organized by Institute of Water Modelling (IWM), Dhaka, Bangladesh (29 September - 09 October 2024).

23-29 June, 2024 — Participated in **2024 Luojia Global Summer School** organized by Wuhan University. Took courses on **Social Sensing and Big Data Computing for Disaster Management** with the instructor Prof Dr. Zhenlong Li from The Pennsylvania State University, USA.

12-18 July, 2023 — Participated in **2023 International Geo-Informatics Summer School** organized by LIESMARS, Wuhan University. Took courses on **Computer Vision and Digital Image Processing** with the instructor Dr. Peng Fu from Harrisburg University, USA.

01-07 July, 2019 — Participated in **2019 Geo-Informatics Summer School** organized by LIESMARS, Wuhan University and Collaborative Innovation Center of Geospatial Technology. Took courses on **Microwave Remote Sensing** with the instructor Prof. Dr. Timo Balz.

2014 — Completed three-month long training program titled **Statistical Computing Certificate Course on MS Excel, SPSS and Programming with R** organized by Department of Statistics, University of Rajshahi, Bangladesh.

Certifications

2025
- **Practical Project Management worked example using Gantt Charts, PMI, PMP, APM** (Udemy)
- **Introduction to Career Skills in Software Development** (LinkedIn)
- **Programming Foundations: Beyond the Fundamentals** (LinkedIn)
- **Microsoft Copilot: The Art of Prompt Writing** (LinkedIn)
- **Programming Foundations: Fundamentals** (LinkedIn)

Certifications (continued)

2024 **AI Python for Beginners: Basics of AI Python Coding** (DeepLearning.AI)

2022 **Learning Data Analytics: 1 Foundations** (LinkedIn)

Communication Foundations (LinkedIn)

Good Citation Behavior (Clarivate)

Data Science for Everyone (DataCamp)

Intermediate Python (DataCamp)

Introduction to Python (DataCamp)

Learning Excel: Data Analysis (LinkedIn)

Writing in the Sciences (Coursera)

Reviewing in the Sciences (Clarivate)

2019 **Workshop on "Hyperspectral image processing"** (ORBITA AEROSPACE)

Research Skills

Remote Sensing & GIS — •Urban land cover classification, impervious surface mapping, and environmental hazard assessment using optical, SAR, atmospheric, and satellite embedding datasets. •Data fusion (Simple Layer Stacking) and texture analysis (GLCM) for accuracy enhancement.

Machine Learning — •Development of novel spectral indices (GISAI, NSAI) and application of ML models (Random Forest, XGBoost, Ensemble Methods, Deep Learning, Neural Networks) for LULC modelling.

Geo-spatial Analysis — •Large-scale dataset management and cloud computing (Google Earth Engine); Geospatial tools (ArcGIS, QGIS, Digital Mapping); programming (Python, R, JavaScript, SQL); temporal data analysis using annual composites.

Research Methodology — •Experimental design for urbanization studies, flood mapping, socio-economic surveys; innovation in addressing spectral confusion and cloud cover; citations, article review, critical thinking, problem solving.

Soft Skills — •Leadership, Project Management (PMP, APM), Teamwork, Public Speaking, Communication, Time Management.

Collaboration — •Interdisciplinary projects with global institutions; peer-reviewed publications in remote sensing and environmental science.

Additional Skills — •Project Management, Gantt Chart, Citations, Article Review, Digital Mapping, Spatial Analysis, Analytical Skills, SEO, Human Behavior, Quantitative Analytics, Spatial Adjustment, Prompt Engineering, Sampling, Spoken English, Spatial Databases, Geospatial Data, Problem Solving, Report Writing, R, Spectral Imaging, GIS, Remote Sensing, Google Earth Engine, Qualitative & Quantitative Research Methodologies, ArcGIS Desktop, IBM SPSS Statistics, JavaScript, Python.

Technical Skills

Programming — Python, JavaScript, R-programming, LaTeX

GIS & RS Platforms — ArcGIS, QGIS, Google Earth Engine (GEE)

Technical Skills (continued)

Geospatial Analytics	Impervious Surface Mapping, Dynamic Land Use Land Cover (LULC) Change Detection, Predictive Modelling, Environmental Hazard Assessment, Risk Mapping (e.g., Flood Mapping & Prediction), Geospatial Data Visualisation & Cartography, Multi-Sensor Data Fusion (Optical, SAR), Advanced Image Processing, Time-Series Data Analysis
AI & ML	Advance Prompt Engineering, Machine Learning Algorithms, Ensemble Learning, Deep Learning Architectures, Cellular Automata,
Data Management	MS Excel, Google Sheets, Power BI, SPSS
Field Surveying	RTK, GPS, Total Station, Learning UAVs for Data Acquisition & Processing

Language Proficiency

Bengali	Native speaker.
English	Professional Command. IELTS test score: overall 6.5.
Dialects	Comprehension of Asian English accents (e.g., Indian English, Chinese English).
Urdu	Conversational proficiency.
Hindi	Conversational proficiency.
Chinese	HSK 1. Taking online course.
Arabic	Can read. Learning online course.

Extracurricular Activities

EGSC **English GeoScience Cafe (EGSC)**, LIESMARS, Wuhan University
- **Editorial Member** (2023–2024): Managed academic content
- **Vice Director** (2017–2018): Organized seminars/conferences
- Coordinated field trips and volunteer programs

SERC **Space & Environment Research Center** (Co-founder)
- **General Secretary** (2014–2015)
- Organized national conference on disaster management

RUPC **Rajshahi University Press Club**
- **Elected Treasurer** (2012–2014)
- **Senior Executive Member** (2014–2015)

RUAA **Rajshahi University Alumni Association**
- Life Member (2025–Present)

References

Md Masud Parves Rana
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Md. Moniruzzaman
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